Rajasthan Board Class 12

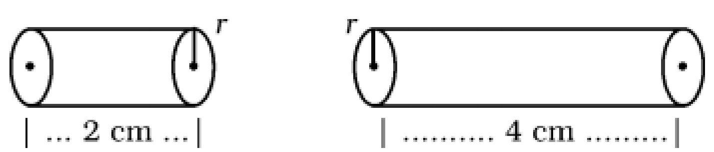
Subject – Physics

Time: $3\frac{1}{4}$ *Hours* **M.M: 56**

1. Write the statement of Gauss's law.
   Sol:
   It states that the total electric flux through a closed surface enclosing a charge is equal to $\frac{1}{\varepsilon_0}$ times the magnitude of the charge enclosed.
   $$\phi = \frac{1}{\varepsilon_0}$$
   $$\Rightarrow \phi = \oint \vec{E} \cdot d\vec{s}$$

2. Calculate the potential at a point due to a charge of $4\times 10^{-9} C$ located $9\times 10^{-2} m$ away from it.
   Sol:
   $$V = \frac{q}{4\pi\varepsilon_0 R}$$
   $$= \frac{4\times 10^{-9}}{9\times 10^{-2} \times 4\times 3.14\times 8.85\times 10^{-12}}$$
   $$= \frac{4\times 10^{-9}}{1000.404\times 10^{-14}}$$
   $$= 0.00399\times 10^{5} V$$

3. In the figure, resistivity's of two conductors of same material are $\rho_1 \Omega - m$ and $\rho_2 \Omega - m$ respectively. Write the value of ratio ρ_1 and ρ_2



4. Write average value of current over a complete cycle of alternating current.
   Sol:
   It is 0.

5. Write the name of electromagnetic wave produced by vacuum tube magnetron.
   Sol:
   Microwaves are produced by Gunn diodes & Magnetrons which are special vacuum tubes

6. Write the relation between object distance (u) image distance (v) and focal length (f) for a concave mirror.

Sol:

$$\frac{2}{R}=\frac{1}{v}+\frac{1}{u}$$

7. (a) Write the name of the lens used to correct near sightedness (myopia).

(b) What will be the radius of curvature of a concave mirror of focal length 10 cm ?

Sol:

(a) Concave lens used to correct near sightedness.

(b) Radius of curvature of the concave mirror is

$$R=2f$$

$$\Rightarrow R=2\times10$$

$$=20cm$$

8. Define stopping potential (cut-off potential) with reference to photoelectric effect.

Sol:

The stopping potential is defined as the potential necessary to stop any electron (or, in other words, to stop even the electron with the most kinetic energy) from 'reaching the other side'.

9. The work function of a metal is $3.31\times1.6\times10^{-19}$ joule. Calculate its threshold frequency in hertz.

Sol:

$$E=h\nu$$

$$=6.626\times10^{-34}\times\nu$$

$$\Rightarrow 3.31\times1.6\times10^{-19}=6.626\times10^{-34}\times\nu$$

$$\Rightarrow \nu=\frac{3.31\times1.6\times10^{-19}}{6.626\times10^{-34}}$$

$$=0.499\times10^{15}s^{-1}$$

10. Write the relation in radius R and mass number A of a nucleus.

Sol:

The relation in radius R and mass number A of a nucleus is :

$$R = R_0A^{1/3}$$

Where,

$$R_0=1.2\times10^{-15}m$$

11. Select one donor impurity among the following :

Boron (B), Aluminium (Al), and Arsenic (As).

Sol:

Arsenic (As) is the donor impurities because having unpaired electrons.

12. A message signal of peak voltage 20 V is used to modulate a carrier wave of peak voltage 30 V. Determine the modulation index.

Sol:

The modulation index

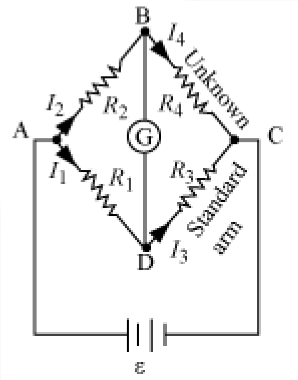
$$\frac{10}{20}$$

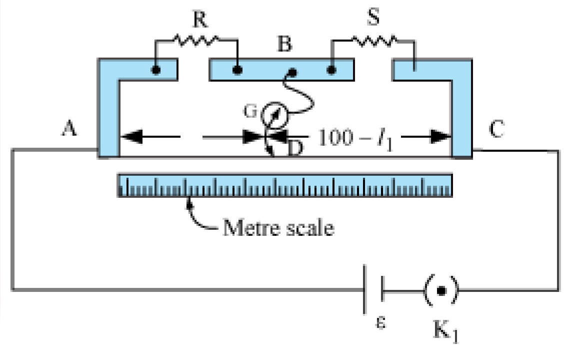
$= 0.5$

13. How the effective power radiated by an antenna will be changed if the wavelength of radiation is decreased?

14. Write the method to determine the value of an unknown resistance by meter bridge and derive necessary formula. Draw circuit diagram.

Sol:





Here,

$AD = l_1$ And $DC = 100 - l_1$

If the resistance per cm is ρ the,

$P = \rho l_1$ and Q=$(100 - l_1)\rho$

So,

$$\frac{Q}{P} = \frac{(100 - l_1)}{l_1}$$

And from the wheat stone bridge

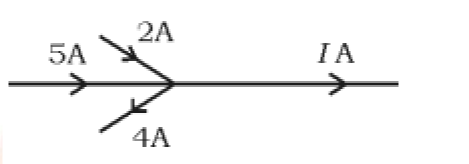
$$\frac{P}{Q}=\frac{R}{S}$$

$$\Rightarrow S=R\frac{Q}{P}=R\frac{(100-l_1)}{l_1}$$

Where, S is the unknown resistance.

15. (a) Define potential gradient.
(b) Write Kirchhoff's junction rule.

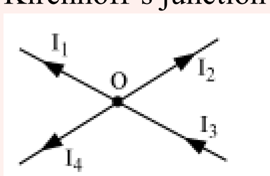
In the given diagram write the value of current I.



Sol:
(a)
Potential gradient is defined as the change in potential with respect to distance.

$$\frac{dV}{dx},\frac{dV}{dy},\frac{dV}{dz}$$

SI unit of potential gradient is $volt$ meter^{-1}
(b) Kirchhoff's junction rule is:



I_1, I_2, I_3 and I_4

Current towards the junction − positive
Current away from the junction − negative

$$I_3+(I_1)+(-I_2)+(-I_4)$$

16. Define angle of declination (magnetic declination). In the magnetic meridian of a certain place, the horizontal component of earth's magnetic field is 0·25 gauss and the dip angle is 60°. At this place find the value of earth's magnetic field.

Sol:
Magnetic declination or variation is the angle on the horizontal plane between magnetic north (the direction the north end of a compass needle points, corresponding to the direction of the Earth's magnetic field lines) and true north (the direction along a meridian towards the geographic North Pole).

Read more on Brainly.in - https://brainly.in/question/2333726#readmore

Earth's magnetic field $H = 0.25$
Angle of dip $\delta = 60^0$
Since,
The horizontal component of the magnetic field is the cosine function of H,
$H = R\cos\delta$

$R = \frac{H}{\cos\delta}$

By putting the values in the upper relation,

$R = \frac{0.25}{\cos 60^0}$

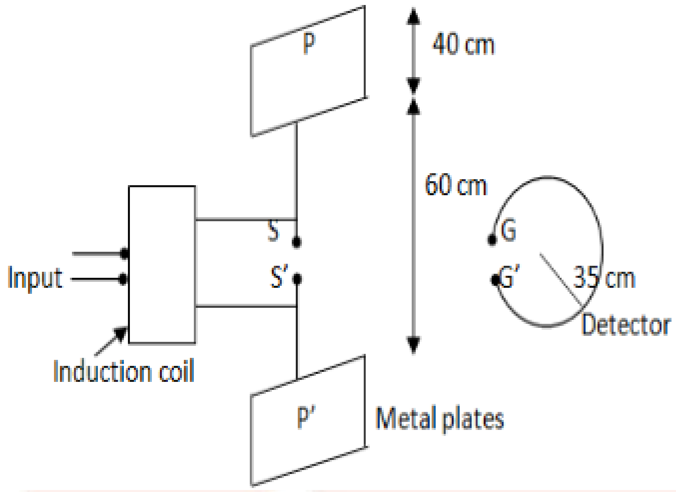
$\Rightarrow R = 0.25 \times 2 = 0.52 units \quad \left[\because \cos 60^0 = \frac{1}{2}\right]$

17. Write the statement of Lenz's law of electromagnetic induction.
A 2 m horizontal long straight conducting wire extending from east to west direction is falling with a speed of 5 m/s perpendicular to the horizontal component of the earth's magnetic field $0.3\times10^{-4} tesla$. Calculate the value of instantaneous emf induced across the ends of wire.
Sol:
Lenz's law states that the induced current produced in a circuit always flows in such a way that it opposes the change or the cause that produces it.
We have
Length of the wire $l = 2cm$,
Falling speed of the wire, $v = 5m/s$
Magnetic field strength is $B = 0.3\times10^{-4} wbm^{-2}$
Therefore,
Emf induced,
$e = Blv$

$= 0.3\times10^{-4} \times 2 \times 5$

$= 0.3\times10^{-5} V$

18. Draw propagation diagram of a linearly polarised electromagnetic wave and write any two properties of electromagnetic waves. The amplitude of the magnetic field associated with an electromagnetic wave in vacuum is $B_0 = 50\times10^{-8} tesla$. Write the value of amplitude of the electric field in V/m associated with the wave.
Sol:



Amplitude of the magnetic field of an electromagnetic wave in vacuum,

$B_0 = 50 \times 10^{-8} tesla$

Speed of light is vacuum is $3 \times 10^8 m/s$

Therefore,

Amplitude of electric field is,

$E = cB_0$

$= 50 \times 10^{-8} \times 3 \times 10^8$

$= 150 N/C$

19. A ray of light is incident at the Brewster's angle on the surface of a transparent medium. Deduce Brewster's law by using Snell's law.

Sol:

Relationship for light waves stating that the maximum polarization (vibration in one plane only) of a ray of light may be achieved by letting the ray fall on a surface of a transparent medium in such a way that the refracted ray makes an angle of 90° with the reflected ray.

According to the Snell’s law,

$$n_{21} = \frac{Speed \text{ of light in midium 1}}{Speed \text{ of light in midium 2}} = \frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

And,

Refractive index of medium1 with respect to medium2 is

$$n_{12} = \frac{Speed \text{ of light in midium 2}}{Speed \text{ of light in midium 1}} = \frac{v_2}{v_1} = \frac{\sin r}{\sin i}$$

$\therefore$

$$n_{21} \times n_{12} = \left(\frac{v_1}{v_2} \times \frac{v_2}{v_1}\right) = \left(\frac{\sin i}{\sin r} \times \frac{\sin r}{\sin i}\right) = 1$$

20. Write de Broglie hypothesis. Obtain the formula for de Broglie wavelength of an electron which is accelerated from rest through a potential V volt.

Sol:

According to de Broglie's hypothesis, a moving material particle sometimes acts as a wave and sometimes as a particle; or a wave is associated with moving material particle, which controls the particle in every respect. The wave associated with moving particle is called matter wave or de Broglie wave, $\lambda = h/(mv)$.

$$L = \frac{h}{mv}$$

where, L is wavelength

$$mv = \frac{h}{L}$$

$$\Rightarrow m^2v^2 = \frac{h^2}{L^2}$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{h^2}{2mL^2}$$

Here, kinetic energy is nothing but the work done by the electron to be accelerated in the electric field out of the potential.

Therefore,

Work done $\frac{h^2}{2mL^2}$

$$Voltage \times charge = \frac{h^2}{2mL^2}$$

$$\Rightarrow V \times e = \frac{h^2}{2mL^2}$$

$$L^2 = \frac{h^2}{2mev}$$

It is the final relationship.

21. Define half-life of a radioactive substance and write the relation of halflife with the following : (a) Radioactive decay constant (disintegration constant)

(b) Mean life of a radioactive substance.

Sol:

The time it takes for half of the radioactive sample that you have to decay. So if you have 100 grams,[with a half-life of 10 years] in 10 years you would have 50 grams, in the next 10 years, 25 more grams would decay.

(a) The decay constant is the fraction of the number of the atoms that decay in one second

$$\frac{dN}{dt} = -\lambda N$$

-ve sign indicates decay

(b) The half life of a radioactive element is the time required for half of it to decay . So if a radioactive element has a half life of one hour, this means that half of it will decay in one hour.

OR

Define mass defect of a nucleus. Binding energy of $_{8}O^{16}$ is 127.5meV. Write the value of its 'binding energy per nucleon'. Write the value of 1 eV energy in joule.

Sol:

The mass defect of nucleus is the change in mass of nucleus with the sum mass of nucleon.

22. On the basis of Bohr's postulates derive an expression for orbital velocity of an electron in n th stationary orbit of hydrogen atom. The ground state energy of hydrogen atom is $(-)XeV$. What will be the kinetic energy of the electron in this state?

Sol:

According to the Bhor's postulat in hydrogen atom, a single electron revolves around a nucleus of charge +e. For an electron moving with a uniform speed in circular orbit on a given radius, the centripetal force is provided by the Coulomb forces of attraction between the electron and the nucleus.

Therefore,

$$\frac{mv^2}{r} = \frac{1(Ze)(e)}{4\pi\varepsilon_0 r^2} ......(1)$$

$$\Rightarrow mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$$

So, K.E is

$$\frac{1}{2}mv^2$$

$$\Rightarrow K.E = \frac{1}{8\pi\varepsilon_0}\frac{Ze^2}{r}$$

Potential energy is given by

$$P.E = \frac{1}{4\pi\varepsilon_0}\frac{(Ze)(-e)}{r} = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$$

Therefore,

Total energy is:

$E = K.E + P.E$

$= \frac{1}{8\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{-1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$

$= \frac{-1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$

For n orbital

$E_n = \frac{-1}{4\pi\varepsilon_0}\frac{Ze^2}{2r}$

The kinetic energy of the electron in this state XeV .

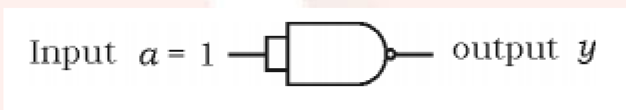
23.

(a) NAND gates are also called universal gate. Why ?

(b) Draw a logic symbol of OR gate.

(c) Write the value of output y in the given circuit

(d)

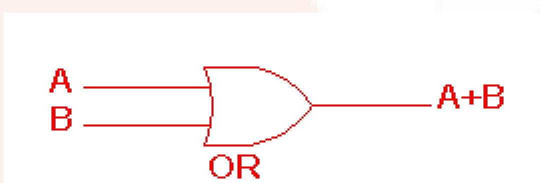


Write the name of the diode used in voltage regulation.

Sol:

(a) The NAND gate and the NOR gate can be said to be universal gates since combinations of them can be used to accomplish any of the basic operations and can thus produce an inverter, an OR gate or an AND gate. The non-inverting gates do not have this versatility since they can't produce an invert.

(b)



(c) The output is 0

Zenor diode is uses in voltage regulation.

24. Explain briefly the following terms used in communication system :

(a) modulation, and (b) transducer.

Sol:

(a) The modulation index (or modulation depth) of a modulation scheme describes by how much the modulated variable of the carrier signal varies around its unmodulated level. It is defined differently in each modulation scheme.

(b) A transducer is a device that converts energy from one form to another. Usually a transducer converts a signal in one form of energy to a signal in another. ... The process of converting one form of energy to another is known as transduction.

25.

	Column I		Column II
i)	Resonant frequency	**a)**	$VI\cos\phi$
ii)	Quality factor	**b)**	$\frac{1}{2}LI^2$
iii)	Average power	**c)**	$\frac{1}{\sqrt{LC}}$
iv)	Impedance	**d)**	$\sqrt{R^2+\left(X_L-X_C\right)^2}$
v)	Magnetic potential energy	**e)**	$\frac{-E}{\left(\frac{dI}{dt}\right)}$
vi)	Coefficient of self-induction	**f)**	$\frac{\omega_0 L}{R}$

26. Define the following :

(a) Total internal reflection

(b) Diffraction of light

(c) Refraction of light.

Sol:

(a) When light is incident from a denser to rarer medium the ray deviates away from the normal. For a certain angle of incidence ($i = i_c$) the angle of refraction becomes 90^0. This angle of incidence is called the critical angle. Any ray with incident angle greater than this critical angle will be completely reflected back to the denser medium. This phenomenon is called Total Internal Reflection.

(b) Diffraction is the slight bending of light as it passes around the edge of an object. The amount of bending depends on the relative size of the wavelength of light to the size of the opening. If the opening is much larger than the light's wavelength, the bending will be almost unnoticeable. However, if the two are closer in size or equal, the amount of bending is considerable, and easily seen with the naked eye.

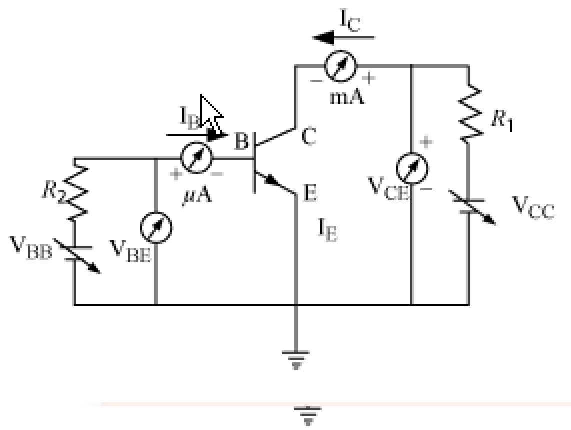
(c) The phenomenon of bending of light from its straight line path as it passes obliquely from one transparent medium to another is called refraction of light. It occurs because the speed of light is different in different media

27. Define common emitter output characteristic for a transistor. Draw a circuit diagram for studying the characteristics of n-p-n transistor in common emitter configuration. Among emitter, base and collector regions of a transistor which one is (a) largest in size and (b) most heavily doped?

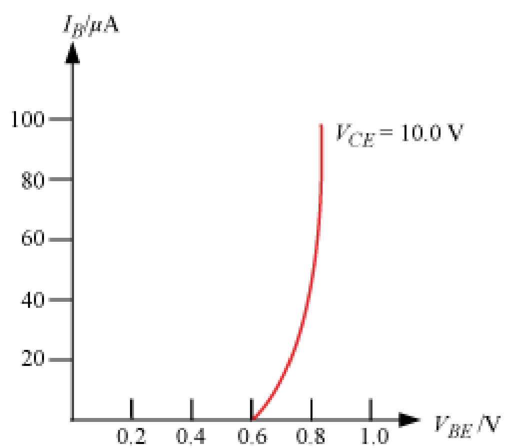
Sol:

In common emitter configuration, input characteristics is the plot obtained by tracing the variation of input current I_B with the input voltage V_{BE}. Similarly, the variation of output current I_C with the Collector to emitter voltage V_{CE} is known as output characteristics.

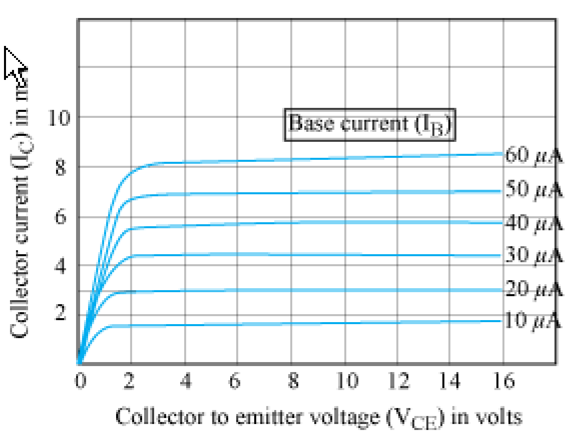
Circuit diagram for tracing characteristic curves of transistor in CE configuration:



Input characteristics



Output characteristics

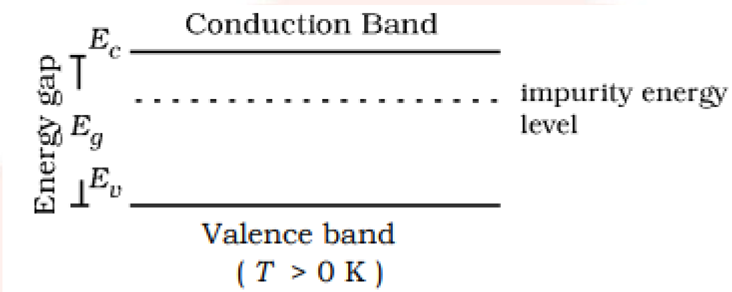


(a) The collector is so named because it collects electrons from base. The collector is the largest of the three regions; it must dissipate more heat than the emitter or base. The transistor has two junctions. ... Because of this the transistor is similar to two diodes, one emitter diode and other collector base diode.

(b) Emitter is heavily doped to provide large no. of majority charge carriers & base and collector are lightly doped to accept these charge charge carriers from emitter.

OR

Define rectification. Draw circuit diagram of a full-wave rectifier. Semi-conductor related to given energy band diagram is : n-type semi-conductor, p-type semi-conductor or intrinsic semiconductor



28. (a) Derive a relation for electric field due to an electric dipole at a point on the equatorial plane of the electric dipole. Draw necessary diagram

(b) An electric dipole of charge $\pm 1\ \mu$ C exists inside a spherical Gaussian surface of radius 1 cm. Write the value of outgoing flux from the Gaussian surface.

(c) Potential on the surface of a charged spherical shell of radius 10 cm is 10 V. Write the value of potential at 5 cm from its center.

Sol:

(a) Let us consider an electric dipole moment $\vec{p} = q(2\vec{a})$

Let P be any point on the equatorial plane of the dipole which is at distance r from the center of the dipole.

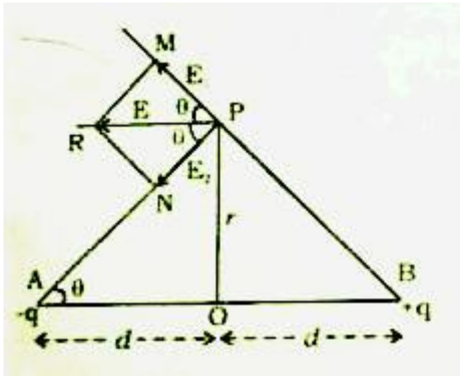
The electric fields due to +q and –q at P is,

$$E_{+q} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2+a^2} \text{ along AP } ....(1)$$

$$E_{-q} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2+a^2} \text{ along BP} ....(2)$$

By resolving this two fields, the components of the electric felds acting on the equatorial line of the dipole cancel each other

Therefore,



$$\vec{E}=\left(E_{+q}+E_{-q}\right)\cos\theta\left(-\hat{P}\right)$$

$$But\left|E_{+q}\right|=\left|E_{-q}\right| \text{ and } \cos\theta=\frac{a}{\sqrt{r^2+a^2}}$$

$$\vec{E}=2E_{+q}\cos\theta\left(-\hat{P}\right)$$

$$\vec{E}=2\cdot\frac{1}{4\pi\varepsilon_0\left(r^2+a^2\right)}\cdot\frac{a}{\sqrt{r^2+a^2}}\left(-\hat{P}\right)$$

$$=\frac{1}{4\pi\varepsilon_0}\frac{2qa}{\left(r^2+a^2\right)^{3/2}}\left(-\hat{P}\right)$$

(b) The value of outgoing flux from the Gaussian surface.

$$\phi=1\times10^{-6}$$

$$\phi=\frac{1\times10^{-6}}{8.8\times10^{-12}}$$

$$=1.1\times10^{5}$$

(c)

$$V=\frac{k}{10}$$

$$k=10v$$

V at r=5

OR

$$=\frac{10V}{5}$$

$$=2V$$

(a) Obtain a relation for equivalent capacitance of the series combination of capacitors. Draw a circuit diagram.

(b) 10 capacitors each of capacity $10\ \mu$ F are joined first in series and then in parallel. Write the value of product of equivalent capacitances.

(c) What will be the value of capacitance of a $4\,\mu$ F capacitor if a dielectric of dielectric constant 2 is inserted fully between the plates of parallel plate capacitor?

Sol:

Let a series combination of three capacitance c_1, v_2 and c_3 are kept across a source of potential. If the amount of charge drawn is q which resides on every capacitor then,

In series,

$$v = v_1 + v_2 + v_3$$

$$v = \frac{q}{c_1} + \frac{q}{c_2} + \frac{q}{c_3}$$

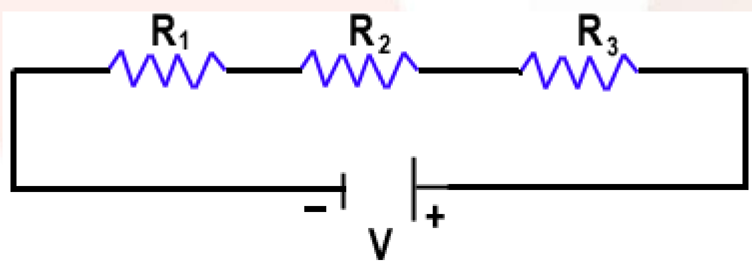
$$v = q\left(\frac{1}{c_1} + \frac{1}{c_2} + \frac{1}{c_3}\right)....(1)$$

If a capacitor of capacitance c draws some amount of charge from the same source then

$$v = \frac{q}{c}$$

$$\frac{q}{c} = q\left(\frac{1}{c_1} + \frac{1}{c_2} + \frac{1}{c_3}\right)$$

$$\frac{1}{c} = \frac{1}{c_1} + \frac{1}{c_2} + \frac{1}{c_3}$$



(b)
Combination of capacitor in series

$$\frac{1}{c_{eq}}=\frac{1}{c_1}+\frac{1}{c_2}+\frac{1}{c_3}+\frac{1}{c_4}+\frac{1}{c_5}+....+\frac{1}{c_{10}}$$

$[\because all$ capacitor having same value$]$

$$\frac{1}{c_{eq}}=\frac{c_1+c_2+c_3+c_4+c_5+c_6+c_7+c_8+c_9+c_{10}}{c_1\cdot c_2\cdot c_3\cdot c_4\cdot c_5\cdot c_6\cdot c_7\cdot c_8\cdot c_9\cdot c_{10}}$$

$$=\frac{(10+10+10+10+10+10+10+10+10+10)10^{-6}}{10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}\cdot10\times10^{-6}}$$

$$=\frac{100\times10^{-6}}{10,000,000,000\times10^{-60}}$$

$$=1\times10^{44}$$

When it connects in parallel

$c_1+c_2+c_3+c_4+c_5+c_6+c_7+c_8+c_9+c_{10}$

$=(10+10+10+10+10+10+10+10+10+10)10^{-6}$

$=100\times10^{-6}$

(c)

29. Write Ampere's circuital law. Obtain an expression for magnetic field on the axis of current carrying very long solenoid. Draw necessary diagram.

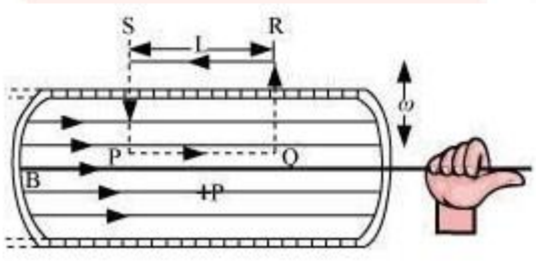
Sol:

Ampère's circuital law relates the integrated magnetic field around a closed loop to the electric current passing through the loop

Definition:

The circulation of the resultant magnetic field along a closed, plane curve is equal to μ_0 times the total current crossing the area bounded by the closed curve provided the electric field inside the loop remains constant. Mathematically it means.

$$\oint B\cdot dl=\mu_0 I$$



Let the magnetic field along the path PQ be B and zero along RS. As the paths QR and PS are perpendicular to the axis of solenoid, the magnetic field component along these paths is zero. Therefore, the path QR and PS will not contribute to the line integral of magnetic field B.

$$\oint\vec{B}\cdot d\vec{l}=\int_P^Q\vec{B}\cdot d\vec{l}+\int_Q^R\vec{B}\cdot d\vec{l}+\int_R^S\vec{B}\cdot d\vec{l}+\int_S^P\vec{B}\cdot d\vec{l}\quad ....(1)$$

Here,

$$\int_P^Q \vec{B}\cdot d\vec{l} = \int_P^Q \vec{B}\cdot d\vec{l}\,\cos 0^0 = Bl$$

Also, consider the solenoid (as $\mathrm{B}=0$)

$$\int \vec{B}\cdot d\vec{l} = 0$$

Therefore,

$$\oint \vec{B}\cdot d\vec{l} = Bl + 0 + 0 + 0 = Bl$$

Now,

From Ampere's Circuital law

$= \mu_0 \times Number$ of turns in rectangle $\times$ Current BL=$\mu_0 nLi$

$\therefore B = \mu_0 nI$

OR

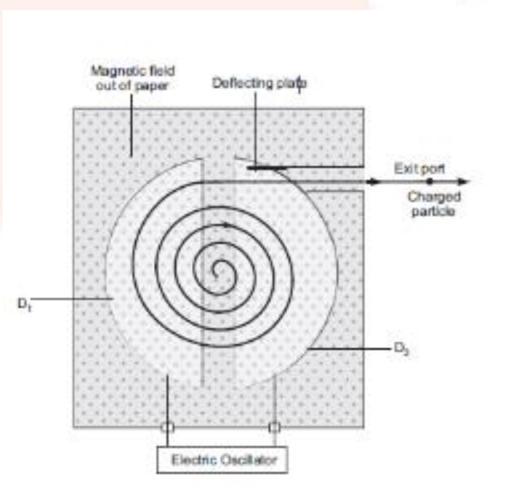
Write the working of cyclotron in brief. Draw a schematic sketch of the cyclotron showing path of accelerated charged particles (ions) in both Dees.

Derive the following parameters of cyclotron :

(a) Cyclotron frequency and

(b) Kinetic energy of ions in cyclotron.

Sol:



(a) The Lorentz force $F_{Lorentz}$ is the centripetal force $F_{Zenrtripetal}$ and causes the particles path to bend in circle.

$$F_{Lorentz} = F_{Zenrtripetal}$$

$$\Rightarrow q \cdot v \cdot B = \frac{mv^2}{r}$$

$$\Rightarrow v = \frac{r \cdot q \cdot B}{m}$$

Where, q is the charge of the particle, v is the velocity of the particle, B is the magnetic flux, m is the mass of the particle r is radius of the circle.

With $v = \omega \cdot r = 2\pi \cdot f \cdot r$

$$2\pi \cdot f \cdot r = \frac{r \cdot q \cdot B}{m}$$

$$\Rightarrow f = \frac{q \cdot B}{2\pi \cdot m}$$

(b)

$$qvB = \frac{mv^2}{r}$$

$$\Rightarrow v = \frac{qBr}{m} .....(1)$$

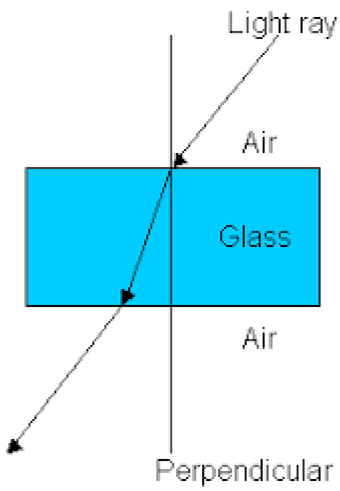
$$K.E = \frac{1}{2} mv^2$$

$$= \frac{q^2 B^2 r^2}{2m} ....(2)$$

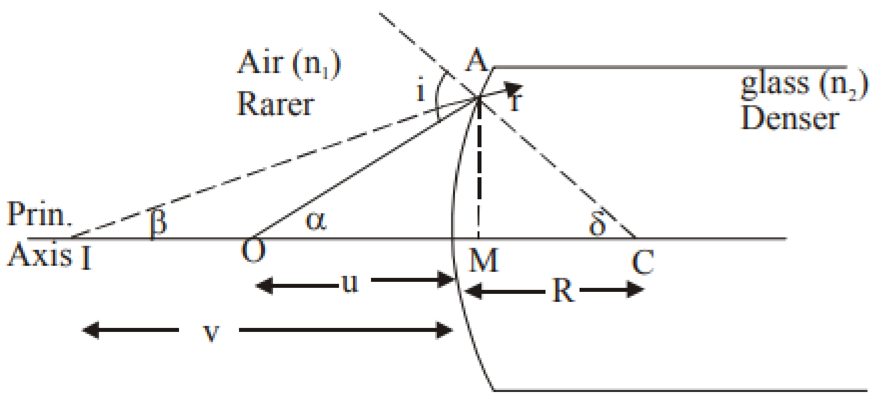
30. Define refraction of light waves. Draw a ray diagram for refraction at a spherical surface separating two media. For refraction at a spherical surface derive the relation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ in object distance (u), image distance (v), refractive index of media (n_1, n_2) and radius of curvature (R).

Sol:

The phenomenon of bending of light from its straight line path as it passes obliquely from one transparent medium to another is called refraction of light. It occurs because the speed of light is different in different media.



Suppose L is a thin lens. The refractive index of the material of lens is n_2 and it is placed in a medium of refractive index n_1. The optical centre of lens is C and X ' X is principal axis. The radii of curvature of the surfaces of the lens are R_1 and R_2 and their poles are P_1 and P_2. The thickness of lens is t, which is very small. O is a point object on the principal axis of the lens. The distance of O from pole P_1 is u. The first refracting surface forms the image of O at I' at a distance v' from P_1.



In ΔOAC

i=α+δ .....(1)

For smaller angle,

$\alpha \simeq \tan a$

$\delta \simeq \tan \delta$

$\therefore i = \tan \alpha + \tan \delta$ .....(2)

In ΔOAM

$$\tan\alpha = \frac{AM}{MO}$$

$But, MO = PO$

$$\therefore \tan\alpha = \frac{AM}{PO} \quad .....(3)$$

$In \Delta ACM$

$$\tan\alpha = \frac{AM}{MC}$$

$But,$ MC=PC

$$\therefore \tan\alpha = \frac{AM}{PC} \quad (4)$$

By using eq (2)

$$i = \left(\frac{AM}{PO} + \frac{AM}{PC}\right) \quad ....(5)$$

$In \ \Delta IAC$

$$r = \beta + \delta \ ...(6)$$

For smaller angle,

α=tanβ

δ=tanδ

$\therefore$ r=taα+tanβ ....(7)

$$r = \left(\frac{AM}{PI} + \frac{AM}{PC}\right) \quad ....7(a)$$

Using Snell's law

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \ ...(8)$$

For smaller angle,

$\sin i \approx i$

$\sin r \approx r$

$$\therefore \frac{i}{r} = \frac{n_2}{n_1}$$

$n_1 i = n_2 r$

Now, using eq(5) And (7a)

$$n_1\left(\frac{AM}{PO}+\frac{AM}{PC}\right)=n_2\left(\frac{AM}{PI}+\frac{AM}{PC}\right)$$

$$\Rightarrow n_1\left(\frac{1}{PO}+\frac{1}{PC}\right)=n_2\left(\frac{1}{PI}+\frac{1}{PC}\right)$$

$$\Rightarrow \frac{n_1}{PO}+\frac{n_1}{PC}=\frac{n_2}{PI}+\frac{n_2}{PC}$$

$$\Rightarrow \frac{-n_2}{PI}+\frac{n_1}{PO}=\frac{n_2}{PC}-\frac{n_1}{PC}$$

$\Rightarrow$

By using proper sign convention

$PI=-v$

$PO=-u$

$PC=R$

We get,

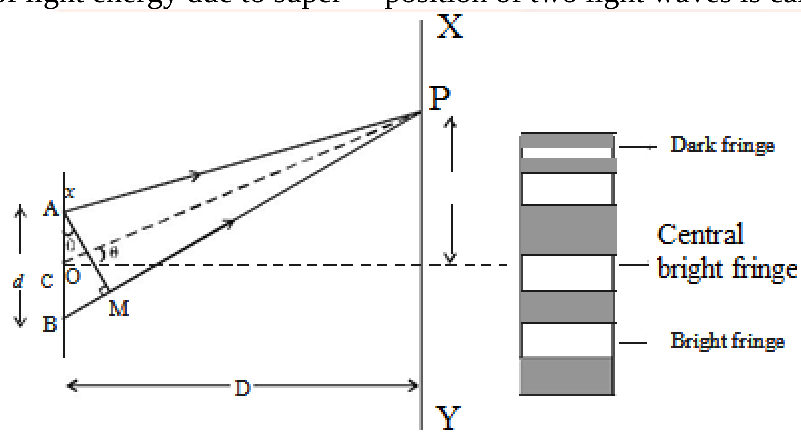
$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$$

OR

Define interference of light waves. Draw a diagram of Young's double silt experiment to produce interference fringe pattern. Derive an expression of fringe width for bright fringes.

Sol:

When two light waves from different coherent sources meet together, then the distribution of energy due to one wave is disturbed by the other. This modification in the distribution of light energy due to super- position of two light waves is called "Interference of light".



In right angled ~~ABM, BM = d sin θ~~ If θ is small,

$\sin\theta = \theta$

Path difference $\delta = \theta \cdot d$

In right angled triangle COP, $\tan\theta = \frac{OP}{CO} = \frac{x}{D}$

For small values of $\theta, \tan\theta = \theta$

Thus, the path difference $\delta = \frac{xd}{D}$

Bright Fringes

By the principle of interference, condition for constructive interference is the path difference is nλ

$\frac{xd}{D}$=nλ

Here,

n=0,1,2......

Which indicates the order of bright fringes

So,

x=(D/d)nλ